

CONCEPTS IN BIG DATA ANALYTICS

COMPREHENSIVE STUDY GUIDE

MODULE 1: INTRODUCTION TO BIG DATA (9 hrs)

1.1 What is Big Data?

- Definition and basic concepts
- Differences between conventional data and big data
- Evolution of Big Data
- **Exam Reference: 2023 Q1, 2024 Q12(a)**

1.2 Big Data Architecture

- Components and layers
- Architectural framework with diagrams
- **Exam Reference: 2024 Q11(a)**

1.3 Big Data Platforms

- Major platforms overview
- Features and capabilities comparison
- **Exam Reference: 2023 Q2, 2024 Q2**

1.4 Types of Data in Big Data

- Structured, unstructured, and semi-structured data
- Examples of each type
- Nature of data and its applications
- **Exam Reference: 2024 Q12(b)**

1.5 Data Analytic Processes and Tools

- Key processes in Big Data analytics
- Common tools used in the industry
- **Exam Reference: 2023 Q12(b)**

1.6 The 5 V's of Big Data

- Volume, Velocity, Variety, Veracity, Value
- Examples for each characteristic

- **Exam Reference: 2023 Q12(a)**

1.7 Big Data Analytical Methods

- Descriptive, diagnostic, predictive, and prescriptive analytics
- Applications and use cases
- Data analysis vs. data reporting
- **Exam Reference: 2023 Q11(a), 2024 Q1**

1.8 Intelligent Data Analysis

- Methods and approaches
- Comparison with traditional analysis
- **Exam Reference: 2023 Q11(b)**

1.9 Big Data Analytics Life Cycle

- Stages and processes
- Implementation considerations
- **Exam Reference: 2024 Q11(b)**

MODULE 2: INTRODUCTION TO STREAM COMPUTING (8 hrs)

2.1 Stream Computing Fundamentals

- Definition and concepts
- Comparison with batch processing
- **Exam Reference: 2023 Q4**

2.2 Streaming Data Architecture

- Components and design
- Implementation considerations
- **Exam Reference: 2023 Q3**

2.3 Stream Data Models

- Characteristics and structure
- Differences from traditional databases

2.4 Sampling in Stream Processing

- Importance and applications
- Various sampling techniques

- **Exam Reference: 2024 Q14(a)**

2.5 Bloom Filters

- Concept and properties
- Working mechanism with examples
- Operations supported by Bloom filters
- **Exam Reference: 2023 Q13(a), 2024 Q13(a)**

2.6 Flajolet-Martin Algorithm

- Purpose and implementation
- Applications in stream data processing
- Finding distinct elements in a stream
- **Exam Reference: 2023 Q13(b), 2024 Q13(b)**

2.7 Moments Estimation in Data Streams

- Concept and importance
- Types of moments
- **Exam Reference: 2023 Q14(b), 2024 Q4**

2.8 DGIM Algorithm

- Purpose and implementation
- Rules for forming buckets
- Applications in stream computing
- Window-based counting
- **Exam Reference: 2023 Q14(a), 2024 Q14(b)**

2.9 Challenges in Stream Processing

- Technical and implementation challenges
- Solutions and approaches
- **Exam Reference: 2024 Q3**

MODULE 3: HADOOP DISTRIBUTED FILE SYSTEM (13 hrs)

3.1 History and Development of Hadoop

- Origins and evolution
- Need for Hadoop in Big Data Analytics
- **Exam Reference: 2023 Q5**

3.2 Hadoop Ecosystem

- Core components
- Interaction between components
- **Exam Reference: 2024 Q16(b)**

3.3 HDFS Architecture

- Design philosophy
- Components and their functions
- Namenodes vs. Datanodes
- **Exam Reference: 2024 Q5**

3.4 HDFS File Storage and Management

- Block storage mechanism
- Replication strategy
- File systems in Hadoop
- **Exam Reference: 2023 Q6**

3.5 HDFS Operations

- Reading files in HDFS
- Writing files in HDFS
- Basic file system commands
- **Exam Reference: 2023 Q15(b), 2024 Q15(b)**

3.6 MapReduce Framework

- Concept and implementation
- Execution pipeline
- Job execution flow
- **Exam Reference: 2023 Q16(a), 2024 Q15(a)**

3.7 Parallel Processing with MapReduce

- How MapReduce enables parallelism
- Runtime coordination and task management
- **Exam Reference: 2023 Q16(b), 2024 Q16(a)**

3.8 MapReduce Examples

- WordCount implementation
- Road Enrichment example
- **Exam Reference: 2023 Q15(a), 2024 Q6**

MODULE 4: PIG, HIVE, AND HBASE (7 hrs)

4.1 Apache Pig

- Introduction and purpose
- Architecture and components
- **Exam Reference: 2024 Q17(a)**

4.2 Pig Execution Modes

- Local mode
- MapReduce mode
- **Exam Reference: 2023 Q7**

4.3 Pig vs. Traditional Databases

- Comparison with SQL
- Advantages and limitations
- **Exam Reference: 2023 Q8, 2024 Q7**

4.4 Grunt and Pig Latin

- Grunt shell usage
- Pig Latin syntax and features
- Data types and operators
- **Exam Reference: 2024 Q18(b)**

4.5 User-Defined Functions in Pig

- Purpose and implementation
- Examples of UDFs

4.6 Hive Architecture

- Components and design
- Services provided by Hive
- **Exam Reference: 2024 Q17(b)**

4.7 Hive Metastore

- Function and importance
- Comparison with traditional databases

4.8 HiveQL and Tables

- Query language features
- Table creation and partitioning
- Types of tables in Hive
- **Exam Reference: 2023 Q17(b), 2024 Q18(a)**

4.9 HBase

- Concepts and purpose
- Architecture and features
- Differences from RDBMS
- Data model and operations
- **Exam Reference: 2023 Q18(b), 2024 Q8**

MODULE 5: INTRODUCTION TO R PROGRAMMING (8 hrs)

5.1 R Programming Fundamentals

- Introduction and importance
- Key features and advantages
- **Exam Reference: 2023 Q9, 2024 Q9**

5.2 R Graphical User Interfaces

- Types and features
- Usage scenarios

5.3 Vectors in R

- Creation and manipulation
- Operations on vectors
- **Exam Reference: 2024 Q19(a)**

5.4 Matrices in R

- Creation methods
- Matrix operations
- Applying functions to rows and columns
- **Exam Reference: 2023 Q20(a)**

5.5 Lists in R

- Definition and usage
- Difference from vectors and matrices
- List operations
- **Exam Reference: 2024 Q20(a)**

5.6 Data Frames in R

- Concept and creation
- Comparison with matrices
- Operations on data frames
- **Exam Reference: 2023 Q19(a), 2024 Q20(b)**

5.7 File Operations in R

- Reading data
- Writing data
- **Exam Reference: 2023 Q19(b)**

5.8 R Programming Examples

- Factorial program
- Fibonacci series program
- Dataset analysis
- **Exam Reference: 2023 Q10, 2024 Q10**

5.9 Data Types and Attributes in R

- Categories and classifications
- Usage in data analysis
- **Exam Reference: 2024 Q19(b)**

EXAM PREPARATION TIPS

1. Focus on frequently asked topics:
 - Bloom filters and stream algorithms
 - HDFS operations
 - MapReduce framework
 - R programming applications
2. Practice programming questions:

- R programs for factorial, Fibonacci series
- MapReduce implementations
- Pig Latin and HiveQL queries

3. Master diagram-based questions:

- Big Data architecture
- HDFS architecture
- Hadoop ecosystem
- Hive and Pig architecture

4. Understand algorithms with examples:

- DGIM algorithm
- Flajolet-Martin algorithm
- Bloom filter operations

5. Prepare short answers for Part A questions:

- Differences between concepts
- Features and characteristics
- Basic definitions and comparisons

This study guide is arranged for optimal comprehension and exam preparation. Use it alongside course materials for best results.

For educational purposes only.